ON-BOARD DIAGNOSIS SYSTEM [START STOP UNIT]

id091400111235

Outline

- The on-board diagnostic function consists of the following functions:
  - — A malfunction detection function, which detects overall malfunctions in the start stop unit-related parts.
  - — A memory function, which stores detected DTCs.
  - — A display function, which indicates malfunction locations and status via DTC output.
  - — A PID/data monitoring function, which reads out specific input/output signals and verifies the input/output condition.
- Using the Mazda Modular Diagnostic System (M-MDS), DTCs can be read out and deleted, and the PID/data monitoring function can be activated.

Malfunction detection function

- Detects malfunctions in input/output signals.
- If a malfunction occurs, the start stop unit records the malfunction as a DTC. A recorded DTC can be read by the Mazda Modular Diagnostic System (M-MDS).

DTC table

×: Applicable
—: Not applicable

DTC No.	KEY warning light (red)	Push button start indicator light (amber)	Description	Fail-safe	Drive cycle	Self test type*1	Memory function
B1026:49	—	Flash	Steering lock malfunction	×	—	C	×
B1026:96	—	Flash	Steering lock malfunction	—	—	C	×
B108B:23	—	Flash	Push button start switch 1 circuit malfunction	×	—	C	×
B108B:24	—	Flash	Push button start switch 1 circuit malfunction	—	—	C	×
B108C:23	—	Flash	Push button start switch 2 circuit malfunction	×	—	C	×
B108C:24	—	Flash	Push button start switch 2 circuit malfunction	—	—	C	×
B10AC:13*2	—	—	Cruise control switch circuit malfunction	—	—	C	×
B10AC:23*2	—	—	Cruise control switch circuit malfunction	—	—	C	×
B10C6:1F*3	On	—	Keyless antenna (exterior, rear) circuit malfunction	×	—	C, D	×
B10C7:1F*3	On	—	Keyless antenna (interior, rear) circuit malfunction	×	—	C, D	×
B10C9:1F	On	—	Keyless antenna (interior, center) circuit malfunction (With keyless entry system)	—	—	C, D	×
	On	—	Keyless antenna (interior, front) circuit malfunction (With advanced keyless entry system)	×	—	C, D	×
B10D1:23*3	On	—	Request switch (LF) circuit malfunction	—	—	C	×
B10D3:23*3	On	—	Request switch (RF) circuit malfunction	—	—	C	×
B10D9:87	—	Flash	Coil antenna (built into push button start) error	—	—	C	—
B10DA:51	—	—	Communication error with PCM (data received failure)	—	—	C	—
B10DA:62	—	—	Communication error with PCM (code mis-matched)	—	—	C	—
B10E7:11	—	Flash	IG1 relay circuit malfunction	—	—	C	×
B10E7:12	—	Flash	IG1 relay circuit malfunction	×	—	C	×
B10E7:16	—	Flash	IG1 relay circuit malfunction	—	—	C	×

DTC No.	KEY warning light (red)	Push button start indicator light (amber)	Description	Fail-safe	Drive cycle	Self test type*1	Memory function
B10E7:17	—	Flash	IG1 relay circuit malfunction	×	—	C	×
B112A:11	—	Flash	IG2 relay circuit malfunction	—	—	C	×
B1140:29*4	—	—	i-stop control error signal received	—	—	C	×
B11FD:1F*3	On	—	Keyless antenna (exterior, LF) circuit malfunction	×	—	C, D	×
B1210:1F*3	On	—	Keyless antenna (exterior, RF) circuit malfunction	×	—	C, D	×
B124C:56	—	—	Turn switch circuit malfunction	—	—	C	×
B13C3:04*3	On	—	LF control unit internal malfunction	×	—	C	×
B13C3:09*3	On	—	LF control unit malfunction	×	—	C	×
B13C3:16*3	On	—	LF control unit power supply voltage low input	×	—	C	×
B13C3:29*3	On	—	Communication error between start stop unit and LF control unit	×	—	C	×
B13D3:05	—	—	Remote transmitter program error	—	—	C	—
B13D3:16	—	—	Low remote transmitter battery voltage	—	—	C	×
B13D3:4A	On	—	Remote transmitter matching malfunction	—	—	C	×
B13D3:51	—	Flash	Unregistered remote transmitter is detected	—	—	C	—
B13D3:94	—	—	Communication error with remote transmitter	—	—	C	—
B13D3:97	—	—	Remote transmitter registration work not completed	—	—	C	×
B13D4:00	On	—	Insufficient remote transmitter programming number	—	—	C	—
B1C53:13	—	—	Windshield wiper switch circuit malfunction	—	—	C	×
C0040:23*5	—	Flash	Brake switch (No.1 signal) circuit malfunction	×	—	C	×
C0040:24*5	—	Flash	Brake switch (No.1 signal) circuit malfunction	×	—	C	×
C0051:2F*6	—	—	Steering angle sensor signal error	—	—	C	×
C0051:95*6	—	—	Steering angle sensor assembly malfunction	—	—	C	×
C0052:14*6	—	—	Steering angle sensor A signal circuit malfunction	—	—	C	×
C0052:2F*6	—	—	Steering angle sensor A signal malfunction	—	—	C	×
C0053:14*6	—	—	Steering angle sensor B signal circuit malfunction	—	—	C	×
C0053:2F*6	—	—	Steering angle sensor B signal malfunction	—	—	C	×
C2003:13*7	—	—	Open circuit in steering switch	—	—	C	×
P0560:16	—	Flash	Start stop unit power supply (ESCL) voltage low input	—	—	C	×
P0615:11	—	Flash	Starter relay circuit malfunction	—	—	C	×
P0615:12	—	Flash	Starter relay circuit malfunction	×	—	C	×
P0615:13	—	Flash	Starter relay circuit malfunction	—	—	C	×

DTC No.	KEY warning light (red)	Push button start indicator light (amber)	Description	Fail-safe	Drive cycle	Self test type*1	Memory function
P0615:16	—	Flash	Starter relay circuit malfunction	—	—	C	×
P0616:23	—	—	Circuit malfunction on downstream side of starter relay	—	—	C	×
P0616:24	—	—	Circuit malfunction on downstream side of starter relay	—	—	C	×
P0616:29	—	—	Downstream side of starter relay signal malfunction	—	—	C	×
P081C:23*5	—	Flash	NOT P position switch circuit malfunction	×	—	C	×
P081C:24*5	—	Flash	NOT P position switch circuit malfunction	×	—	C	×
P081D:29*8	—	Flash	Neutral switch error signal received	×	—	C	×
P081D:2F*8	—	Flash	Neutral switch signal mismatch	×	—	C	×
P0830:23*8	—	Flash	CPP switch circuit malfunction	×	—	C	×
P0850:23*5	—	Flash	Transaxle range sensor circuit malfunction	×	—	C	×
P0850:24*5	—	Flash	Transaxle range sensor circuit malfunction	×	—	C	×
P0930:71*5	—	—	Shift-lock solenoid circuit malfunction	×	—	C	×
P1708:24*8	—	Flash	Starter interlock switch circuit malfunction	×	—	C	×
P1708:29*8	—	Flash	Starter interlock switch circuit or CPP switch circuit malfunction	×	—	C	×
P1794:16	—	Flash	Start stop unit power supply voltage (terminal 2W) low input	—	—	C	×
U0001:88	On	Flash	Module communication error (HS-CAN)	×	—	C	×
U0028:87	On	—	Communication error with BCM	—	—	C	×
U0100:00	On	Flash	Communication error with PCM	×	—	C	×
U0100:87	—	—	Communication error with PCM (no response or data mis-match)	—	—	C	—
U0101:00*5	—	Flash	Communication error with TCM	×	—	C	×
U0121:00	On	Flash	Communication error with ABS HU/CM or DSC HU/CM	×	—	C	×
U0121:87	On	Flash	Communication error with ABS HU/CM or DSC HU/CM	×	—	C	×
U0131:00	—	—	Communication error with EPS control module	×	—	C	×
U0140:00	—	—	Communication error with BCM	×	—	C	×
U0146:00	On	Flash	Communication error with BCM	×	—	C	×
U0151:00	On	Flash	Communication error with SAS control module	×	—	C	×
U0155:00	On	Flash	Communication error with instrument cluster	×	—	C	×
U0447:86	On	Flash	Error signal received from BCM	×	—	C	×
U201F:11	On	—	Keyless receiver circuit malfunction	—	—	C	×
U201F:12	On	—	Keyless receiver circuit malfunction	—	—	C	×

DTC No.	KEY warning light (red)	Push button start indicator light (amber)	Description	Fail-safe	Drive cycle	Self test type*1	Memory function
U2100:00	On	—	Start stop unit configuration not completed	×	—	C	×
U2300:00	—	—	Configuration data not received from instrument cluster	×	—	C	×
U3000:01*6	—	—	Start stop unit internal malfunction	—	—	C	×
U3000:49	On	Flash	Start stop unit internal malfunction	×	—	C	×
U3000:96	On	Flash	Start stop unit malfunction	×	—	C	×
	On	Flash	Start stop unit malfunction	—	—	C	×
U3003:16	On	—	Start stop unit power supply voltage (terminal 1A) low input	×	—	C	×
U3004:11	—	Flash	ACC relay circuit malfunction	—	—	C	×
U3004:12	—	Flash	ACC relay circuit malfunction	×	—	C	×
U3004:16	—	—	ACC relay circuit malfunction	—	—	C	×
U3004:17	—	—	ACC relay circuit malfunction	×	—	C	×

*1 : C: CMDTC self test, D: ODDTC self test
*2 : With cruise control system
*3 : With advanced keyless entry system
*4 : With i-stop system
*5 : ATX
*6 : With smart city brake support (SCBS)
*7 : With steering switch
*8 : MTX

Snapshot Data

Note
- The start stop unit stores the following two types of snapshot data (vehicle information) when a DTC is detected and displays them in the Mazda Modular Diagnostic System (M-MDS).
  - — Vehicle information detected by start stop unit
  - — Vehicle information detected by instrument cluster and received by start stop unit via CAN communication

- The data for all DTCs currently detected is stored.

—: Not applicable

Snapshot data item	Unit		Data contents	Data read/use method	Corresponding data monitor items
AAT	°C	°F	Ambient temperature	—	—
APP_STATUS	Accelerator Pedal Off/Under20%/ Over20%/FAIL		Accelerator pedal position status	—	—
CFG_STATUS	Config Complete/ Not Configured/ Config Error		Instrument cluster configuration status	—	—
ECT_STATUS	Under 0 degrees C/ 0 - Under 80 degrees C/Over 80 degrees C/FAIL		Engine coolant temperature status	—	—

Snapshot data item	Unit	Data contents	Data read/use method	Corresponding data monitor items
IC_VPWR	V	Instrument cluster power supply voltage	• The start stop unit constantly receives the power supply voltage value of the instrument cluster sent via CAN communication from the instrument cluster. • If a DTC is detected, the start stop unit records the power supply voltage of the instrument cluster when the DTC was detected, and it is displayed in the Mazda Modular Diagnostic System (M-MDS).	VPWR*1
IG-ON_TIMER	hh:mm:ss*2	Elapsed time since ignition was switched ON (engine off or on) **Note** • The instrument cluster records the elapsed time since the ignition was switched ON (engine off or on).	• The start stop unit constantly receives the elapsed time since the ignition was switched ON (engine off or on) sent via CAN communication from the instrument cluster. • If a DTC is detected, the start stop unit records the elapsed time since the ignition was switched ON (engine off or on) when the DTC was detected, and it is displayed in the Mazda Modular Diagnostic System (M-MDS).	—
PWR_MODE_KEY	Key Out/Key Recently Out (Position 0)/ Accessory (Position 1)/Post Ignition (Position 2)/Ignition On (Position 2)/ Running (Position 2)/Running - Starting In Progress (Position 2)	• Key Out: Ignition switched off • Key Recently Out (Position 0): Elapsed time **within 3 s** since ignition was switched off • Accessory (Position 1): Ignition is switched to ACC • Post Ignition (Position 2): Elapsed time **within 3 s** since ignition was switched ON (engine off or on) • Ignition On (Position 2): Ignition switched ON (engine off) • Running (Position 2): Ignition switched ON (engine on) • Running - Starting In Progress (Position 2): Cranking condition	• The start stop unit constantly receives the ignition switch status sent via CAN communication from the instrument cluster. • If a DTC is detected, the start stop unit records the ignition switch status when the DTC was detected, and it is displayed in the Mazda Modular Diagnostic System (M-MDS).	—
RPM_STATUS	Engine Stop/ Under1500rpm/ Over1500rpm/FAIL	Engine speed status	• The start stop unit constantly receives the engine speed sent via CAN communication from the instrument cluster. • If a DTC is detected, the start stop unit records the engine speed when the DTC was detected, and it is displayed in the Mazda Modular Diagnostic System (M-MDS).	TACHOMTR*1

Snapshot data item	Unit		Data contents	Data read/use method	Corresponding data monitor items
SHIFT_STATUS	P/N/D/R/FAIL		Selector lever position status	• The start stop unit constantly receives the selector lever position sent via CAN communication from the instrument cluster. • If a DTC is detected, the start stop unit records the selector lever position when the DTC was detected, and it is displayed in the Mazda Modular Diagnostic System (M-MDS).	—
TOTAL_DIST	km	Miles	Accumulated total traveled distance from completion of vehicle until start stop unit detects DTC (Odometer value in instrument cluster)	The total traveled distance from which the start stop unit detects DTCs to the present can be calculated by performing the following procedure. 1. Verify the odometer value in the instrument cluster. 2. Verify the snapshot data item TOTAL_DIST. 3. Subtract 2 from 1.	—
TOTAL_TIME	hh:mm:ss*2		Accumulated total elapsed time since vehicle completion until start stop unit detects a DTC **Note** • When the ROOM fuse is removed, and the ignition is switched off, the time is not included in the elapsed time.	The elapsed time from which the start stop unit detects DTCs to the present can be calculated by performing the following procedure. 1. Verify the instrument cluster PID item TOTAL_TIME. 2. Verify the snapshot data item TOTAL_TIME. 3. Subtract 2 from 1.	TOTAL_TIME*1

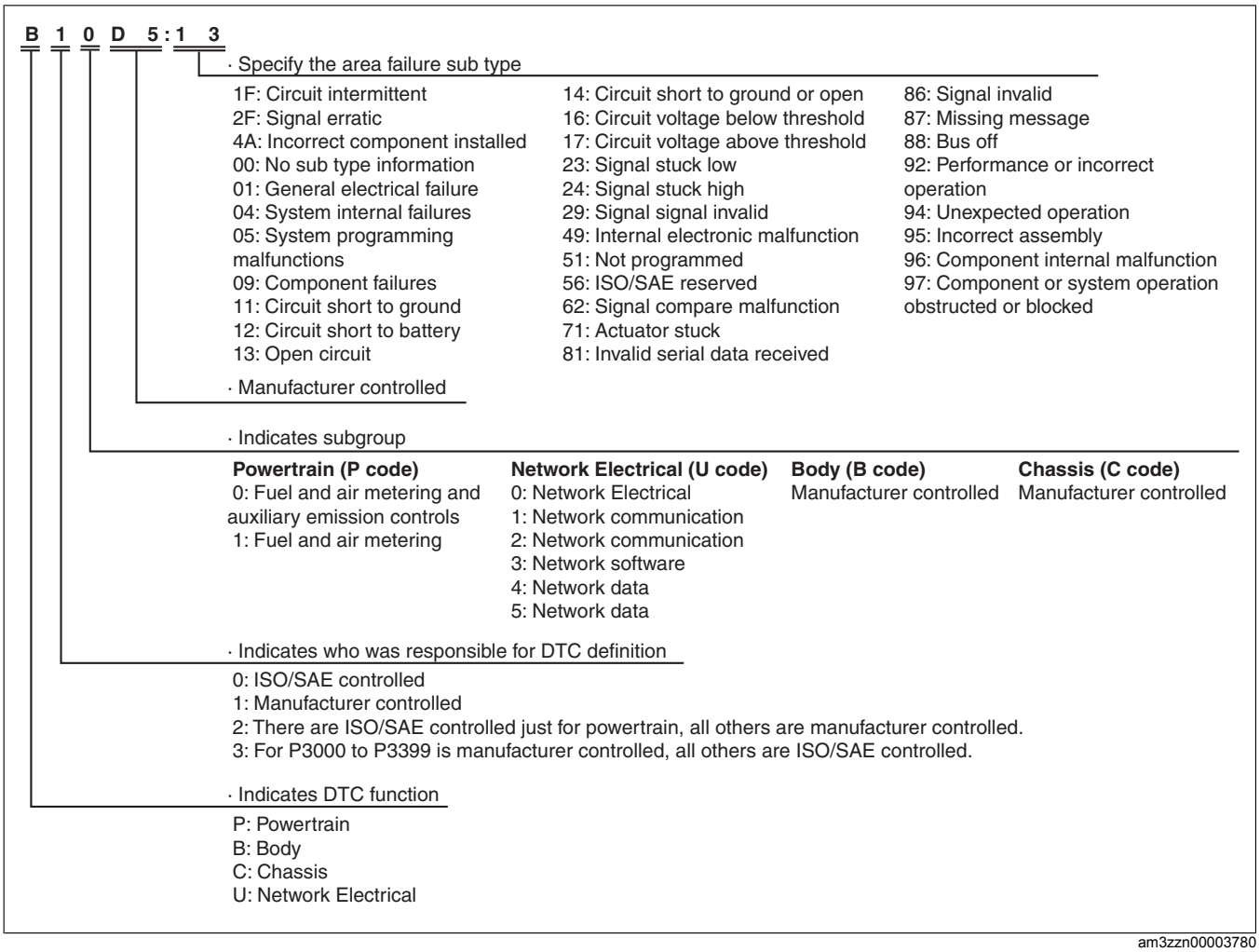
Snapshot data item	Unit	Data contents	Data read/use method	Corresponding data monitor items
TRNS_NUM_L	Initial/No.1/No.2/No.3/No.4/No.5/No.6	Registration number of the remote transmitter	If the start stop unit detects DTC B13D3:16 (Low remote transmitter battery voltage), inspect the remote transmitter in which the malfunction is occurring using the following procedure to display the registration number of the remote transmitter that was operated. 1. Verify the registration number of the remote transmitter in the snapshot data. 2. Display the start stop unit PID item RF_TRNS_NUM and inspect the battery voltage of the remote transmitter having the same registration number as the recorded remote transmitter registration number.	—
VPWR	V	Start stop unit power supply voltage	—	VPWR_B2
VPWR_B1	V	Start stop unit power supply voltage (ROOM fuse)	—	VPWR_B1
VSPD_STATUS	Stop/0-10km/h/Over10km/h/FAIL	Vehicle speed status	• The start stop unit constantly receives the vehicle speed sent via CAN communication from the instrument cluster. • If a DTC is detected, the start stop unit records the vehicle speed when the DTC was detected, and it is displayed in the Mazda Modular Diagnostic System (M-MDS).	SPEEDOMTR*1

*1 : Instrument cluster PID (See ON-BOARD DIAGNOSTIC [INSTRUMENT CLUSTER].)

*2 : The seconds may be indicated after the decimal point.

DTC 7-digit code definition

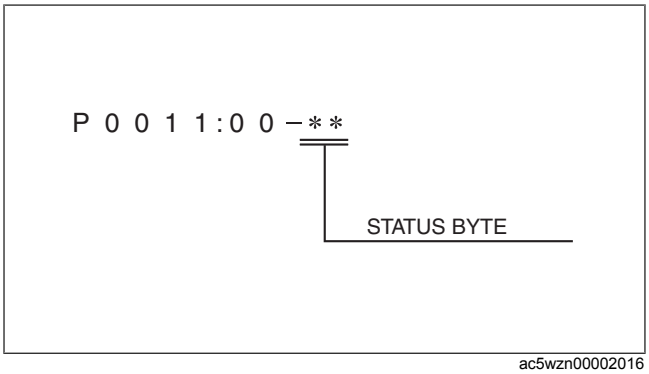
- When related systems or components have failed, the CM stores the DTC of the malfunctioning part in the CM memory, and allows for the retrieval of the store data using scanning tool when necessary. The DTCs are indicated using seven digits. Each digit indicates the following.



am3zzn00003780

Status byte for DTC

- The two digits (two digits after hyphen (-)) after the seven-digit DTC.
- A code which indicates the pending code, current/ past malfunction status, or warning illumination status.
- Can be read by performing a CMDTC self-test using the Mazda Modular Diagnostic System (M-MDS).
- For details on the status byte, refer to the explanation on the Mazda Modular Diagnostic System (M-MDS) when reading the DTC.



ac5wzn00002016

Detection condition for the applicable DTC

DTC No.	Description	Detection condition
B1026:49	Steering lock malfunction	• Start stop unit detects malfunction in the steering lock of the internal start stop unit.
B1026:96	Steering lock malfunction	• Start stop unit detects malfunction in the steering lock of the internal start stop unit.
B108B:23	Push button start switch 1 circuit malfunction	• Start stop unit detects that the push button start switch 1 is stuck on for **120 s or more**.
B108B:24	Push button start switch 1 circuit malfunction	• Start stop unit detects that the push button start switch 1 is kept in an off condition **5 times or more** even though the push button start switch 2 is on.
B108C:23	Push button start switch 2 circuit malfunction	• Start stop unit detects the push button start switch 2 is stuck on for **120 s or more**.

DTC No.	Description	Detection condition
B108C:24	Push button start switch 2 circuit malfunction	• Start stop unit detects that the push button start switch 2 is kept in an off condition **5 times or more** even though the push button start switch 1 is on.
B10AC:13	Cruise control switch circuit malfunction	• Start stop unit detected an open circuit in the cruise control switch circuit for **5 s or more** with the ignition switched ON (engine off or on).
B10AC:23	Cruise control switch circuit malfunction	• Start stop unit detects any cruise control switch stuck on for **120 s or more** with the ignition switched ON (engine off or on).
B10C9:1F	Keyless antenna (interior, center) circuit malfunction	• Start stop unit detects an open circuit or short to ground in the keyless antenna (interior, center) with the ignition switched off.
B10E7:11	IG1 relay circuit malfunction	• Start stop unit detects the voltage from the IG1 relay output monitor as **less than 2.5 V** for **0.5 s or more** while the engine is not being cranked with the ignition switched ON (engine off or on).
B10E7:12	IG1 relay circuit malfunction	• Start stop unit detects the IG1 relay output monitor voltage of **2.5 V or more** for **0.5 s or more** with the ignition switched off or ACC.
B10E7:16	IG1 relay circuit malfunction	• Start stop unit detects the IG1 monitor voltage of **less than 2.5 V** for **1 s or more** while the engine is not being cranked with the ignition switched ON (engine off or on).
B10E7:17	IG1 relay circuit malfunction	• Start stop unit detects the IG1 monitor voltage of **2.5 V or more** for **1 s or more** with the ignition switched off or ACC.
B112A:11	IG2 relay circuit malfunction	• Start stop unit detects the IG2 relay output monitor voltage of **2.5 V or less** for **0.5 s or more** with the ignition switched ON (engine off or on).
B1140:29	i-stop control error signal received	• Start stop unit detects that the starter ground circuit does not turn on or open circuit on even though the PCM detects an engine restart request while the engine is stopped under i-stop control for **2 s or more** for a total of **5 times or more**.
B124C:56	Turn switch circuit malfunction	• Start stop unit detects that the turn switch LH and RH are turned on simultaneously for **5 s or more**.
B13D3:4A	Remote transmitter matching malfunction	• Start stop unit detects a remote transmitter different from the one equipped to the vehicle.
B13D3:97	Remote transmitter registration work not completed	• Start stop unit detects that the registration of a remote transmitter has not been completed.
B1C53:13	Windshield wiper switch circuit malfunction	• Start stop unit detects an open circuit, short to ground, or short to power supply in the windshield wiper switch circuit for **5 s or more** with the ignition switched ON (engine off or on).
C0040:23	Brake switch (No.1 signal) circuit malfunction	• Start stop unit detects that the following signals are unmatched for 0.5 s or more for a continuous 5 times. — Transaxle range sensor (ATX) or starter interlock switch (MTX) condition — Starter relay downstream circuit condition — PCM control result — CAN signal condition output from PCM
C0040:24	Brake switch (No.1 signal) circuit malfunction	• The start stop unit detected the following condition **5 times** continuously with the ignition switched ON (engine off or on). — The brake switch (No.1 signal) is ON for a continuous **3 s or more** even though the brake switch (No.2 signal) changes.
C0051:2F	Steering angle sensor signal error	• Start stop unit detects the steering angle sensor signal error for **5 s or more** with the ignition switched ON (engine off or on).

DTC No.	Description	Detection condition
C0051:95	Steering angle sensor assembly malfunction	• Start stop unit receives the following signals with the ignition switched ON (engine off or on). — Steering angle sensor assembly malfunction from EPS control module for **1 s or more** — Improper assembly of steering angle sensor at **360°** — Steering wheel angle of **720° or more** for **0.048 s or more**
C0052:14	Steering angle sensor A signal circuit malfunction	• Start stop unit detects an open circuit or short to ground in the steering angle sensor A signal circuit.
C0052:2F	Steering angle sensor A signal malfunction	• When any of the following conditions is met: — Start stop unit detected the sudden change in steering angle sensor A signal. — Start stop unit detected that the steering angle sensor A signal is stuck with the ignition switched ON (engine off or on).
C0053:14	Steering angle sensor B signal circuit malfunction	• Start stop unit detects an open circuit or short to ground in steering angle sensor B signal circuit.
C0053:2F	Steering angle sensor B signal malfunction	• When any of the following conditions is met: — Start stop unit detects the sudden change in steering angle sensor B signal. — Start stop unit detects that the steering angle sensor B signal is stuck with the ignition switched ON (engine off or on).
C2003:13	Open circuit in steering switch	• Start stop unit detects an open circuit in the steering switch circuit for **5 s or more** with the ignition switched ON (engine off or on).
P0560:16	Start stop unit power supply (ESCL) voltage low input	• Start stop unit power supply (ESCL) circuit voltage of **less than 8.5 V** is detected for **5 s or more**.
P0615:11	Starter relay circuit malfunction	• Start stop unit detects that the starter relay monitor circuit voltage of **less than 2.2 V** for **0.2 s or more** with the ignition switched off.
P0615:12	Starter relay circuit malfunction	• Start stop unit detects that the starter relay circuit voltage of specification or more for **0.1 s or more** with the ignition switched off.
P0615:13	Starter relay circuit malfunction	• Start stop unit detects an open circuit in the starter relay circuit for **0.075 s or more** with the ignition switched off.
P0615:16	Starter relay circuit malfunction	• Start stop unit detects that the starter relay circuit voltage that is less than the specification for **0.05 s or more** during cranking.
P0616:23	Circuit malfunction on downstream side of starter relay	• Start stop unit detects a short to ground in the starter relay downstream circuit continuously for **0.5 s or more** for a continuous **5 times**.
P0616:24	Circuit malfunction on downstream side of starter relay	• Start stop unit detects an open circuit in the starter relay downstream circuit continuously for **0.5 s or more** for a continuous **5 times**.
P0616:29	Downstream side of starter relay signal malfunction	• Start stop unit detects that the following signals are unmatched for **0.5 s or more** for a continuous **5 times**. — Transaxle range sensor (ATX) or starter interlock switch (MTX) condition — Starter relay downstream circuit condition — PCM control result — CAN signal condition output from PCM
P081C:23	NOT P position switch circuit malfunction	• Start stop unit detects that the NOT P position switch is stuck on for **5 s or more** for a continuous **5 times** with the ignition switched ON (engine off or on).
P081C:24	NOT P position switch circuit malfunction	• Start stop unit detects that the NOT P position switch is stuck off for **5 s or more** for a continuous **5 times** with the ignition switched ON (engine off or on).
P081D:29	Neutral switch error signal received	• Start stop unit receives the neutral switch signal error signal from the PCM for **1 min or more** with the ignition switched ON (engine off or on).

DTC No.	Description	Detection condition
P081D:2F	Neutral switch signal mismatch	• Start stop unit detects that the vehicle speed changes from **0 km/h {0 mph}** to **10 km/h {6.2 mph}** for a continuous **5 times or more** even though the neutral signal from the PCM is being received with the ignition switched ON (engine off or on).
P0830:23	CPP switch circuit malfunction	• Start stop unit detects that the CPP switch is stuck on for **10 min or more**.
P0850:23	Transaxle range sensor circuit malfunction	• The start stop unit detected the following condition **5 times** continuously with the ignition switched ON (engine off or on). — A transaxle range sensor signal is in the P or N position for a continuous **1 s or more** even though the TCM CAN signal is in a position other than the P and N positions.
P0850:24	Transaxle range sensor circuit malfunction	• The start stop unit detected the following condition **5 times** continuously with the ignition switched ON (engine off or on). — A transaxle range sensor signal is in a position other than the P and N positions for a continuous **1 s or more** even though the TCM CAN signal is in the P or N position.
P0930:71	Shift-lock solenoid circuit malfunction	• Start stop unit detects the shift-lock solenoid is stuck on for **30 min or more** with the ignition switched ON (engine off or on).
P1708:24	Starter interlock switch circuit malfunction	• Start stop unit detects an open circuit in the starter interlock switch circuit for **0.5 s or more** for a continuous **5 times** with the ignition switched ON (engine off or on).
P1708:29	Starter interlock switch circuit or CPP switch circuit malfunction	• Start stop unit detects that the starter interlock switch circuit is stuck on for **0.2 s or more**, or the CPP switch circuit is stuck off for **0.25 s or more**.
P1794:16	Start stop unit power supply voltage (terminal 2W) low input	• Start stop unit power supply circuit voltage (terminal 2W) of **less than 6.5 V** is detected for **5 s or more**.
U0001:88	Module communication error (HS-CAN)	• Start stop unit detects the CAN bus communication line (HS-CAN) malfunction **10 times** continuously with the ignition switched ON (engine off or on).
U0028:87	Communication error with BCM	• Start stop unit detects the communication error with BCM **10 times continuously**.
U0100:00	Communication error with PCM	• Start stop unit could not receive the CAN signal from the PCM for **1 s or more** with the ignition switched ON (engine off or on).
U0101:00	Communication error with TCM	• Start stop unit could not receive the CAN signal from the TCM for **1 s or more** with the ignition switched ON (engine off or on).
U0121:00	Communication error with ABS HU/CM or DSC HU/CM	• Start stop unit could not receive the CAN signal from the ABS HU/CM or DSC HU/CM for **1 s or more** with the ignition switched ON (engine off or on).
U0121:87	Communication error with ABS HU/CM or DSC HU/CM	• Start stop unit could not receive the CAN signal from the ABS HU/CM or DSC HU/CM for **10 s or more** with the ignition switched off.
U0131:00	Communication error with EPS control module	• Start stop unit could not receive the CAN signal from the EPS control module for **1.2 s or more** with the ignition switched ON (engine off or on).
U0140:00	Communication error with BCM	• Start stop unit could not receive the CAN signal from the BCM for **5 s or more** with the ignition switched ON (engine off or on).
U0146:00	Communication error with BCM	• Start stop unit could not receive the CAN signal from the BCM for **5 s or more** with the ignition switched ON (engine off or on).
U0151:00	Communication error with SAS control module	• Start stop unit could not receive the CAN signal from the SAS control module for **1 s or more** with the ignition switched ON (engine off or on).

DTC No.	Description	Detection condition
U0155:00	Communication error with instrument cluster	• Start stop unit could not receive the CAN signal from the instrument cluster for **5 s or more** with the ignition switched ON (engine off or on).
U0447:86	Error signal received from BCM	• Start stop unit receives the error signals from the BCM for **5 s or more** with the ignition switched ON (engine off or on).
U201F:11	Keyless receiver circuit malfunction	• Start stop unit detects that the keyless receiver circuit voltage of **less than 3.4 V** for **0.5 s or more**.
U201F:12	Keyless receiver circuit malfunction	• Start stop unit detects that the keyless receiver circuit voltage of **6.2 V or more** for **0.5 s or more**.
U2100:00	Start stop unit configuration not completed	• Start stop unit configuration not completed.
U2300:00	Configuration data not received from instrument cluster	• Start stop unit cannot receive the configuration data from the instrument cluster for **1 s or more** with the ignition switched ON (engine off or on).
U3000:01	Start stop unit internal malfunction	• Start stop unit detects the internal malfunction.
U3000:49	Start stop unit internal malfunction	• Start stop unit detects the internal malfunction.
U3000:96	Start stop unit malfunction	• Start stop unit malfunction detected.
U3003:16	Start stop unit power supply voltage (terminal 1A) low input	• Start stop unit power supply circuit voltage (terminal 1A) of **5V or more, less than 8.5 V** is detected for **5 s or more**.
U3004:11	ACC relay circuit malfunction	• Start stop unit detects that the ACC relay output monitor voltage of **less than specification** for **0.05 s or more** with the ignition switched to ACC or ON (engine off or on).
U3004:12	ACC relay circuit malfunction	• Start stop unit detects that the ACC relay output monitor voltage of **2.5 V or more** for **0.5 s or more** with the ignition switched off.
U3004:16	ACC relay circuit malfunction	• Start stop unit detects that the ACC monitor voltage of **less than 2.5 V** for **1 s or more** with the ignition switched to ACC or ON (engine off or on).
U3004:17	ACC relay circuit malfunction	• Start stop unit detects that the ACC monitor voltage of **2.5 V or more** for **5 s or more** with the ignition switched off.

Data Monitor Function

- With the PID/data monitor function, input/output signal monitor items set in the start stop unit can be selected and read out in real-time.
- The items which can be selected are as follows.

—: Not applicable

PID	Unit/ Operation	Data contents	Data read/use method	Module control terminal
BRAKE_SW1*1	Off/On	Brake switch (No.1 signal) signal input status • Off: Brake switch (No.1 signal) is off. • On: Brake switch (No.1 signal) is on.	—	1C (Brake switch)
CLUT_CUT_SW*2	Off/On	Starter interlock switch input status • Off: Starter interlock switch is off. • On: Starter interlock switch is on.	—	1F (Starter interlock switch)
CLUTCH_SW*2	Off/On	Clutch switch input status • Off: CPP switch is off. • On: CPP switch is on.	—	2E (Clutch pedal position switch)
COMM_ST_TSP	Off/On	Displays operation condition of the immobilizer system key code read function • Off: Key code read function not performed • On: Code read function is performed and coil antenna reads key code from remote transmitter transponder.	Key code reading function of immobilizer system is performed and when a brake switch, clutch switch, or push button start signal is input, the coil antenna reads the key code from the transponder of the remote transmitter, and displays On while the key code is being sent to the start stop unit.	• 1AC (PATS+) • 1AE (PATS-)

PID	Unit/ Operation	Data contents	Data read/use method	Module control terminal
CRU_CON_SW*3	Not_Pressed/On/Off/ On/Off/ Cancel/ Resume/ Fault/Set (+)/Set (-)/ Distance (+)/ Distance (-)	Cruise control switch operation condition • Not_Pressed: Each switch of cruise control switch is not pressed or speed limiter switch is pressed. • On: Cruise ON switch is pressed. • Off: Cruise OFF switch is pressed. • On/Off: — • Cancel: CANCEL switch is pressed. • Resume: RESUME (RES) switch is pressed. • Fault: Cruise control switch has a malfunction. • Set (+): SET (+) switch is pressed. • Set (-): SET (-) switch is pressed. • Distance (+): — • Distance (-): —	—	1Y (Cruise control switch signal)
ES_UL_SW	Not_Unlock/ Unlock	Steering lock unlock switch condition • Not_Unlock: Steering lock unlock switch is off. • Unlock: Steering lock unlock switch is on.	—	—
ES_UL_SW_M	Not_Unlock/ Unlock	Memory value of steering lock unlock switch • Not_Unlock: Memory value of steering lock unlock switch is not unlocked. • Unlock: Memory value of steering lock unlock switch is unlocked.	—	—
ES_ULL_SW	Off/On	Steering lock unlock/lock switch condition • Off: Steering lock unlock/lock switch is off. • On: Steering lock unlock/lock switch is on.	—	—
ES_ULL_SW_M	Off/On	Memory value of steering lock unlock/lock switch • Off: Memory value of steering lock unlock/lock switch is off. • On: Memory value of steering lock unlock/lock switch is on.	—	—
F_FOG_SW	Off/On	Front fog light switch operation condition • Off: Fog light switch is not in F.FOG position. • On: Fog light switch is in F.FOG position.	—	—
H/L_SW_HI	Off/On	Headlight switch (HI) operation condition • Off: Light switch is not in HI position. • On: Light switch is in HI position.	—	—
H/L_SW_LOW	Off/On	Displays dimmer switch position of the light switch that is being input to the start stop unit • Off: Dimmer switch of light switch is not in LO position • On: Dimmer switch of light switch is in LO position	Verify if the monitor value switches when the dimmer switch of the light switch is switched between LO and HI positions. If the dimmer switch operation and monitor values do not match, inspect the light switch.	—
H/L_SW_OFF	Off/On	Headlight light switch (OFF) operation condition • Off: Light switch is not in OFF position. • On: Light switch is in OFF position.	—	—

PID	Unit/ Operation	Data contents	Data read/use method	Module control terminal
H/L_SW_PASS	Off/On	Headlight switch (PASS) operation condition • Off: Light switch is not in passing position. • On: Light switch is in passing position.	—	—
H/L_SW_TNS	Off/On	Headlight switch (TNS) operation condition • Off: Light switch is not in TNS position. • On: Light switch is in TNS position.	—	—
HAZARD_SW	Off/On	Displays condition of hazard warning switch input to start stop unit • Off: Hazard warning switch is off. • On: Hazard warning switch is on.	—	1G (Hazard warning switch signal)
IG_POSITION	OFF/ACC/ ON/CRANK	Displays power status controlled by the start stop unit • OFF: Start stop unit controls ignition switch off • ACC: Start stop unit controls ignition switch ACC • ON: Start stop unit controls ignition switch ON (engine off) • CRANK: Start stop unit controls cranking at ignition switch ON	• When the push button start is operated, the start stop unit performs power source switching and displays the PID value of the power source position in which control was performed. • If the push button start operation and PID value do not match, and a power source status of another unit differs from the PID value, inspect the input power source of the other unit because the power source control of the start stop unit is normal.	—
INFO_SW*4	Off/ SW1_On/ SW2_On/ SW3_On/ Unknown/ Invalid	Steering switch operation condition • Off: Steering switch UP/DOWN/INFO switch is not pressed. • SW1_On: UP switch is pressed. • SW2_On: DOWN switch is pressed. • SW3_On: INFO switch is pressed. • Unknown: Steering switch UP/DOWN/ INFO switch signal is not determined. • Invalid: Steering switch malfunction is received.	Displays INFO switch position of steering switch input to start stop unit. Steering switch signal is input to start stop unit via clock spring.	• 1AA (Steering switch signal) • 1AB (Steering switch ground)
INHIBIT_SW*1	Off/On	Transaxle range switch input status • Off: Selector lever is not in P and N positions. • On: Selector lever is in P or N position.	—	1F (Selector lever position switch)
LG/T_LK_SW*5	Off/On	Request switch (liftgate) operation condition • Off: Request switch (liftgate) is off. • On: Request switch (liftgate) is on.	—	2F (LF control unit communication)
LG/T_UNL_SW	Off/On	**4SD** • Trunk lid opener switch operation condition — Off: Trunk lid opener switch is off. — On: Trunk lid opener switch is on. **5HB** • Liftgate opener switch operation condition — Off: Liftgate opener switch is off. — On: Liftgate opener switch is on.	—	1Q (BCM communication)

PID	Unit/ Operation	Data contents	Data read/use method	Module control terminal
LL_SW_D_LK	Off/On	Driver's door lock link switch (lock side) operation condition • Off: Front door lock-link switch (driver's side) is in unlock. • On: Front door lock-link switch (driver's side) is in lock.	—	1P (Lock input)
NUM_TRNSMIT	—	Displays number of the remote transmitters programmed to the start stop unit	—	—
P_RANGE_SW*1	Off/On	Not P position switch condition • Off: NOT P position switch is off. • On: NOT P position switch is on.	—	1J (Not P position switch)
PAD_DOWN_SW*6	Off/On	Steering shift switch (DOWN) condition • Off: DOWN switch of steering shift switch is off. • On: DOWN switch of steering shift switch is on.	—	1N (Steering shift switch (down))
PAD_UP_SW*6	Off/On	Steering shift switch (UP) condition • Off: UP switch of steering shift switch is off. • On: UP switch of steering shift switch is on.	—	1L (Steering shift switch (up))
R_FOG_SW	Off/On	Rear fog light switch condition • Off: Fog light switch is not in R.FOG position. • On: Fog light switch is in R.FOG position.	—	—
RELAY_ACC	Off/On	ACC relay operation condition • Off: ACC relay drive is off. • On: ACC relay drive is on.	—	2C (ACC relay control)
RELAY_IG1	Off/On	IG1 relay operation condition • Off: IG1 relay drive is off. • On: IG1 relay drive is on.	—	2S (IG1 relay control)
RELAY_IG2	Off/On	IG2 relay operation condition • Off: IG2 relay drive is off. • On: IG2 relay drive is on.	—	2D (IG2 relay control)
RF_RECEP_ST	Initial/ Transmitter/ RF_Response	Displays radio reception status from the remote transmitter • Initial: Signal from remote transmitter is not received by start stop unit • Transmitter: Signal by button operation on remote transmitter is received by start stop unit • RF_Response: During remote transmitter authorization, response signal from remote transmitter is received by start stop unit	When verifying the PID value while operating the remote transmitter buttons, if “Transmitter” is displayed it can be determined that the transmission between the remote transmitter, keyless receiver, and start stop unit is normal.	2R (Keyless receiver communication)

PID	Unit/ Operation	Data contents	Data read/use method	Module control terminal
RF_TRNS_NUM	Initial/No.1/ No.2/No.3/ No.4/No.5/ No.6	When pressing the unlock button of the remote transmitter, the registration number of the remote transmitter sending radio waves to the start stop unit is displayed. • Initial: Registration number of remote transmitter cannot be received • No.1: Registration number No.1 remote transmitter is received • No.2: Registration number No.2 remote transmitter is received • No.3: Registration number No.3 remote transmitter is received • No.4: Registration number No.4 remote transmitter is received • No.5: Registration number No.5 remote transmitter is received • No.6: Registration number No.6 remote transmitter is received	When DTC B13D3:16 (Low remote transmitter battery voltage) is detected, the registration number of the remote transmitter in which the battery voltage has decreased is displayed in the snapshot data item TRNS_NUM_L. Display the registration number of the remote transmitter using the PID, verify the remote transmitter with the low battery voltage, and inspect the battery condition of the target remote transmitter.	2R (Keyless receiver communication)
RQ_SW_LF*5	Off/On	Request switch (LF) operation condition • Off: Request switch (LF) is off. • On: Request switch (LF) is on.	The signal is input to the start stop unit through the LF control unit of the request switch (LF).	2F (LF control unit communication)
RQ_SW_RF*5	Off/On	Request switch (RF) operation condition • Off: Request switch (RF) is off. • On: Request switch (RF) is on.	The signal is input to the start stop unit through the LF control unit of the request switch (RF).	2F (LF control unit communication)
SECURITY_I	Off/On	Output condition of security indicator • Off: Security indicator light is turned off. • On: Security indicator light is illuminated.	Security indicator on request to instrument cluster	2M, 2O (CAN)
SFT_LK_SOL*1	Off (Lock)/ On (Unlock)	Shift lock solenoid operation condition • Off (Lock): Shift-lock is operated. • On (Unlock): Shift-lock is released.	—	2G (Shift lock)
SPD_LIM_SW*7	Off/On/ Unknown	Speed limiter switch input status • Off: Speed limiter switch is off. • On: Speed limiter switch is on. • Unknown: Speed limiter switch signal is not determined.	—	1Y (Cruise control switch signal)
SSB_1	Off/On	Push button start switch 1 input signal • Off: Push button start switch 1 is off. • On: Push button start switch 1 is on.	—	1H (Push button start 1)
SSB_2	Off/On	Push button start switch 2 input signal • Off: Push button start switch 2 is off. • On: Push button start switch 2 is on.	—	2J (Push button start 2)
SSB_AMBER	Off/On	Push button start indicator light (amber) illumination request • Off: Push button start indicator light (amber) is turned off. • On: Push button start indicator light (amber) is illuminated.	—	1M (Push button illumination (amber))
SSB_GREEN	Off/On	Push button start indicator light (green) illumination request • Off: Push button start indicator light (green) is turned off. • On: Push button start indicator light (green) is illuminated.	—	1K (Push button illumination (green))
SSB_ILLMI	Off/On	Push button start illumination request • Off: START/STOP/ENGINE indicator light is turned off. • On: START/STOP/ENGINE indicator light is illuminated.	—	1I (Push button illumination)
START_RLY_1	V	Displays voltage at start stop unit terminal 2V	—	2V (Starter relay)

PID	Unit/ Operation	Data contents	Data read/use method	Module control terminal
START_RLY_2	V	Displays voltage at start stop unit terminal 1D	—	1D (Starter monitor)
STR_A_ANG_A*8	° (deg)	Displays steering angle sensor A signal aiming angle	—	—
STR_A_ANG_B*8	° (deg)	Displays steering angle sensor B signal aiming angle	—	—
STR_AB_ANG*8	° (deg)	Displays steering angle signal (absolute angle) • Steering wheel in neutral position: Near **0 degrees** • Steering wheel turned to left: Changes from **0 degrees** to positive • Steering wheel turned to right: Changes from **0 degrees** to negative	—	—
STR_ANG_S*8	Without/ With	Vehicle with/without steering angle sensor • Without: Steering angle sensor is not equipped. • With: Steering angle sensor is equipped.	—	—
STR_ANG_S_A*8	V	Displays voltage at the steering angle sensor A signal	—	1U (Steering angle sensor (signal A/ D1)) **Note** • Refer to the [STEERING ANGLE SENSOR INSPECTION] in the workshop manual.
STR_ANG_S_B*8	V	Displays voltage at the steering angle sensor B signal	—	1W (Steering angle sensor (signal A/ D1)) **Note** • Refer to the [STEERING ANGLE SENSOR INSPECTION] in the workshop manual.
STR_ANG_S_V*8	V	Displays steering angle sensor power supply voltage **Note** • “5 V” is displayed even if the voltage is **5 V or more**.	—	1S (Steering angle sensor power supply)
T_SEN_LF	Off/On	Displays in the Mazda Modular Diagnostic System (M-MDS) but it does not operate.		
T_SEN_RF	Off/On	Displays in the Mazda Modular Diagnostic System (M-MDS) but it does not operate.		
TURN_SW_L	Off/On	Turn switch (LH) input • Off: Turn switch is in position other than LH. • On: Turn switch is in LH position.	—	—
TURN_SW_R	Off/On	Turn switch (RH) input • Off: Turn switch is in position other than RH. • On: Turn switch is in RH position.	—	—

PID	Unit/ Operation	Data contents	Data read/use method	Module control terminal
VPWR_ACC	V	Displays ACC power supply voltage input to start stop unit	If there is a malfunction in the ACC power supply voltage input to the start stop unit, the keyless entry system and push button start system do not operate normally because PID item VPWR_ACC is used for ACC power on/off determination.	1E (ACC monitor)
VPWR_B1	V	Displays B+ power supply (ROOM fuse) voltage input to start stop unit	If there is a malfunction in the B+ power supply (ROOM fuse) voltage input to the start stop unit, the keyless entry system does not operate normally because PID item VPWR_B1 is used for determining whether or not there is a ROOM fuse inserted.	1A (Power supply)
VPWR_B2	V	Displays B+ power supply (AT fuse) voltage input to start stop unit	When the ROOM fuse is removed, the B+ power supply (AT fuse) is the start stop unit power supply. If there is a malfunction in the B+ power supply (AT fuse) voltage input to the start stop unit when the ROOM fuse is removed, the push button start system does not operate normally.	2W (Power supply)
VPWR_B3	V	Displays B+ power supply (SRS2/ESCL fuse) input to start stop unit.	If there is a malfunction in the B+ power supply (SRS2/ESCL fuse) voltage input to the start stop unit, the steering lock and the push button start system do not operate normally because PID item VPWR_B3 is used as the steering lock power supply built into the start stop unit.	2X (Power supply)
VPWR_IG1	V	Displays IG1 power supply voltage input to start stop unit	If there is a malfunction in the IG1 power supply voltage input to the start stop unit, the keyless entry system and the push button start system do not operate normally because PID item VPWR_IG1 is used for IG1 power on/off determination.	1B (Power supply)
WASHER_F	Off/On	Windshield washer switch signal input status • Off: Windshield washer switch is in OFF position. • On: Windshield washer switch is in ON position.	—	—

PID	Unit/ Operation	Data contents	Data read/use method	Module control terminal
WASHER_R	Off/On	Rear washer switch signal input status • Off: Rear washer switch is not in ON position. • On: Rear washer switch is in ON position.	—	—
WIP_F_INT	Off/On	Windshield wiper switch (INT) signal input status • Off: Windshield wiper switch is in MIST, OFF or LO position. • On: Windshield wiper switch is in AUTO*9/INT*10 or HI position.	—	—
WIP_F_LOW	Off/On	Windshield wiper switch (LO) signal input status • Off: Windshield wiper switch is in OFF or AUTO*9/INT*10 position. • On: Windshield wiper switch is in MIST, LO or HI position.	—	—
WIP_INT_E_A	Low/High	Wiper INT encoder A signal input status • Low: Windshield wiper switch INT encoder A value is low. • High: Windshield wiper switch INT encoder A value is high.	—	—
WIP_INT_E_B	Low/High	Wiper INT encoder B signal input status • Low: Windshield wiper switch INT encoder B value is low. • High: Windshield wiper switch INT encoder B value is high.	—	—
WIP_INT_E_C	Low/High	Wiper INT encoder C signal input status • Low: Windshield wiper switch INT encoder C value is low. • High: Windshield wiper switch INT encoder C value is high.	—	—
WIP_R_INT	Off/On	Rear wiper switch (INT) signal input status • Off: Rear wiper switch is not in INT position. • On: Rear wiper switch is in INT position.	—	—
WIP_R_ON	Off/On	Rear wiper switch (ON) signal input status • Off: Rear wiper switch is not in ON position. • On: Rear wiper switch is in ON position.	—	—

*1 : ATX
*2 : MTX
*3 : With cruise control system
*4 : With steering switch
*5 : With advanced keyless entry system
*6 : With steering shift switch
*7 : With adjustable speed limiter (ASL)
*8 : With smart city brake support (SCBS)
*9 : With auto wiper system
*10 : Without auto wiper system